

SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE

Department of Computer Science Engineering

Course Code: 23CS002PC304

Course Title: AI Assisted Coding

Year/Sem: III / II

Regulation: R23

Assignment Number: 2.1 / 24

Week: 1 – Monday

Name: O.Narasimha Raju

HT No: 2303A53034

Batch: 46

Lab Objectives

- Explore AI coding tools such as Gemini, GitHub Copilot, and Cursor
- Compare AI-generated solutions
- Understand correctness and efficiency of generated code
- Learn debugging and improvement of AI code

Task 1: Armstrong Number Check

Python Code

```
num = int(input("Enter number: "))
temp = num
sum = 0

while temp > 0:
    digit = temp % 10
    sum += digit ** 3
    temp //= 10
```

```

temp //= 10

if sum == num:
    print("Armstrong Number")
else:
    print("Not Armstrong Number")

```

Explanation: Checks whether a number is equal to the sum of cubes of its digits.

Task 2: Leap Year Check

Python Code

```

year = int(input("Enter year: "))

if (year % 4 == 0 and year % 100 != 0) or (year % 400 == 0):
    print("Leap Year")
else:
    print("Not Leap Year")

```

Explanation: Uses standard leap year conditions.

Task 3: Comparison of AI Tools

- Gemini provides detailed explanations
- GitHub Copilot provides real-time suggestions
- Cursor helps in editing and modifying entire code blocks
- Copilot is faster for coding, Gemini is better for learning

Task 4: Refactoring Code

Improved Armstrong Code

```

def is_armstrong(num):
    digits = str(num)
    power = len(digits)
    total = sum(int(d)**power for d in digits)
    return total == num

```

```
n = int(input())
if is_armstrong(n):
    print("Armstrong")
else:
    print("Not Armstrong")
```

Explanation: Improved readability and flexibility using functions.

Conclusion

AI coding tools improve development speed and productivity.

However, developers must validate logic, optimize code, and ensure correctness before using in real applications.